Top DSA Questions in Java

Master these topics before your next interview

@java_in.depth

ARRAYS & TWO POINTERS

Must-know patterns

Q1 Find the two numbers that sum to a target (Two Sum)

Q2 Rotate an array by k steps in-place

Q3 Find the maximum subarray sum (Kadane's Algorithm)

Q4 Merge two sorted arrays without extra space

STRINGS

Commonly asked in FAANG

Q5 Check if a string is a palindrome

Q6 Longest substring without repeating characters

Q7 Group anagrams from a list of strings

Q8 Find the minimum window substring containing all chars

LINKED LISTS

Classic pointer problems

Q9 Reverse a singly linked list

Q10 Detect a cycle in linked list (Floyd's algorithm)

Q11 Find the middle node using slow & fast pointers

Q12 Merge K sorted linked lists efficiently

TREES & BST

Recursion is your friend

Q13 Implement inorder, preorder, postorder traversal

Q14 Find the Lowest Common Ancestor of two nodes

Q15 Check if a binary tree is height balanced

Q16 Serialize and deserialize a binary tree

DYNAMIC PROGRAMMING

The boss-level topic

Q17 Fibonacci — memoization vs tabulation

Q18 0/1 Knapsack problem using DP table

Q19 Longest Common Subsequence (LCS)

Q20 Edit Distance between two strings

GRAPHS

BFS · DFS · Dijkstra

Q21 BFS vs DFS — when to use which traversal?

Q22 Detect a cycle in a directed graph

Q23 Topological sort of a DAG

Q24 Shortest path using Dijkstra's algorithm

SORTING & SEARCHING

Know your complexities

Q25 implement QuickSort and explain its worst case

Q26 Binary search on a rotated sorted array

Q27 Find the kth largest element using a heap

Q28 Count inversions in an array (merge sort based)

STACK & QUEUE

Know your complexities

Q29 implement a stack using two queues

Q30 Validate balanced parentheses using a stack

Q31 Find the next greater element for each array item

Q32 Largest rectangle in a histogram

You're one step closer to the offer!

Practice these 32 questions consistently.
Focus on time & space complexity in Java.
Save this carousel for your daily revision.

SAVE & FOLLOW FOR MORE →